Case F

Mr. Robert Miller, 64M — Day 2 Post-CABG (Median Sternotomy) | 30 pack-year smoking · Mild baseline COPD ·

Target: 70%+ | Cardiorespiratory Physiotherapy

■ PROBLEM LIST

- Post-sternotomy pain → splinting → basal atelectasis
- Reduced chest wall compliance post-surgery
- Shallow apical breathing (impaired diaphragm recruitment)
- Weak, guarded, ineffective cough
- Mild baseline COPD (pre-existing airflow obstruction)
- Sternal instability / "clicking" — requires characterisation
- Reluctant to mobilise, breathe deeply, or cough
- Unable to clear secretions effectively
- Bed-bound → DVT risk, deconditioning
- Fear-avoidance behaviour limiting recovery participation
- Mobility limited by chest drain and telemetry attachments

■ RED FLAGS

None identified

■ PATHOPHYSIOLOGY

- Surgical mechanism of PPC: Median sternotomy divides the sternum and retracts the chest wall, causing direct thoracic cage trauma.
- Atelectasis cascade: Surfactant dysfunction from bypass, diaphragmatic splinting from pain, and impaired cough reflex allow distal airways to collapse.
- Gravitational forces compound basal zone collapse in supine positioning.
- The result is ventilation/perfusion (V/Q) mismatch, hypoxaemia, and a fertile environment for post-operative pneumonia.
- COPD additive risk: Robert's mild COPD means he has reduced mucociliary clearance, increased mucus production, and already-compromised small airway function at baseline.

■ DIFFERENTIAL DIAGNOSES

1. Post-operative Pneumonia

- Common PPC following cardiac surgery (incidence 2–22%)
- Presents with productive cough, fever, CXR consolidation, raised WCC
- May develop from retained secretions, aspiration during intubation, or haematogenous spread

2. Pulmonary Embolism (PE)

- Post-surgical hypercoagulable state + immobility = significantly elevated DVT/PE risk
- Presents with pleuritic chest pain, tachycardia, dyspnoea, and sudden SpO■ drop
- Sternal pain could mask pleuritic PE pain in this patient

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Chest Expansion Measurement

What it measures Bilateral chest wall excursion; identifies basal restriction due to pain and atelectasis; quantifies improvement with treatment How to perform Patient seated or semi-reclined. Tape measure at 4th intercostal space (mid-chest) and at xiphisternum (lower chest). Re

Incentive Spirometry Assessment

What it measures Sustained maximal inspiratory effort; degree of atelectasis and inspiratory muscle weakness; visual biofeedback tool How to perform Patient in high Fowler's (60–80°). Apply sternal support pillow first. Instruct slow maximal inhalation through IS mouthpiece to ra

Sternal Stability Assessment

What it measures Integrity of sternal wiring post-sternotomy; determines safety to proceed with coughing, upper limb exercise, and progressive mobilisation How to perform Observe wound for visible gap, haematoma, or discharge. Palpate gently around sternal edges for crepitus, mob

■ TREATMENT PLAN

■ Initial Phase — Days 1–7 (ICU Step-down / Cardiac Ward)

- Sternal precautions (8–12 weeks): No pushing or pulling with arms — no steering wheel pushing, no lifting from the floor. Always hug the pillow when coughing. If the sternum feels unstable or clicking worsens, stop and call the nurse immediately.
- Deep breathing is medicine: Shallow breathing is the main cause of lung complications after heart surgery. Ten deep breaths with the incentive spirometer every hour while awake is one of the most important things Robert can do to prevent pneumonia.
- Smoking cessation: This is the single most impactful change Robert can make post-CABG — cessation reduces re-infarction risk, improves wound healing, restores ciliary function, and prolongs graft patency. A formal cessation referral will be arranged.
- Cardiac rehabilitation: A structured 6–8 week outpatient cardiac rehab programme significantly reduces mortality and improves quality of life after CABG — Robert will be referred before discharge.

■ Long-term — Weeks 2–12 (Outpatient / Cardiac Rehab)

- Cardiac Rehabilitation (Phase II & III): Supervised aerobic exercise (walking, cycling) at 60–80% HR reserve; telemetry initially; 6–8 week structured programme with individualized progression
- COPD management: Spirometry review at 6–8 weeks post-discharge; optimise inhaler therapy; smoking cessation programme ongoing with follow-up
- Home exercise programme: Daily walking target starting at 10–15 min, building to ≥30 min; sternal precaution education reinforced until 12 weeks post-surgery
- Discharge criteria: Ambulating ≥200 m independently without desaturation · Pain controlled on oral analgesia only · Sternal wound healed and stable · Independently performing breathing exercises · Cardiac rehab referral confirmed · Smoking cessation support engaged